Native AI Bootcamp

Lab Guide - Autonomous AI

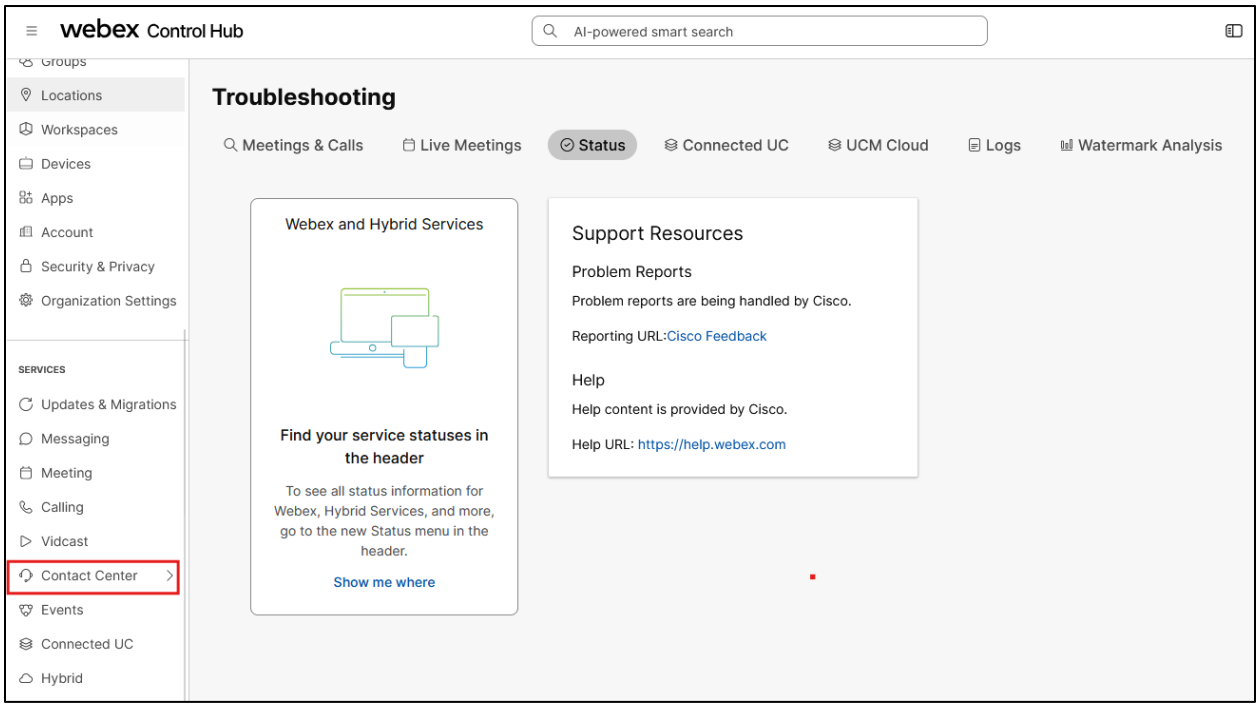
Cisco TAC

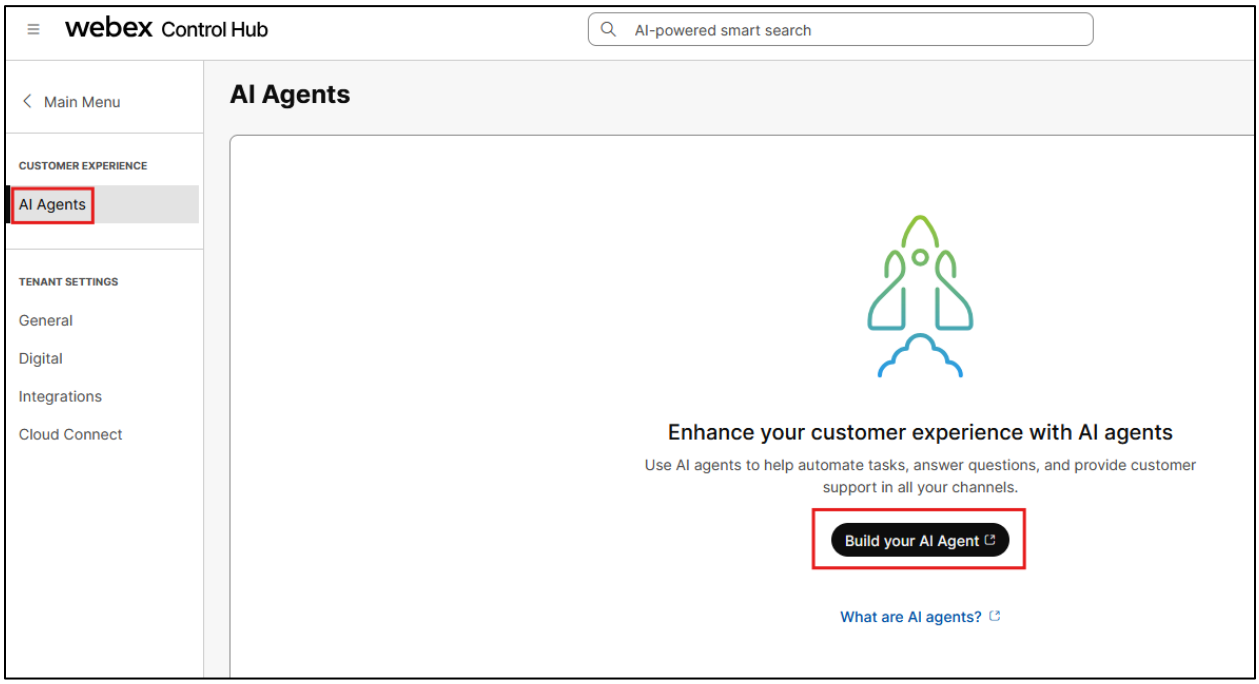
March 2026

Lab Link: ***https://expo.ciscodcloud.com/sedqzuzordovfhi43sem46dj***

Lab 1: Create a Knowledge Base

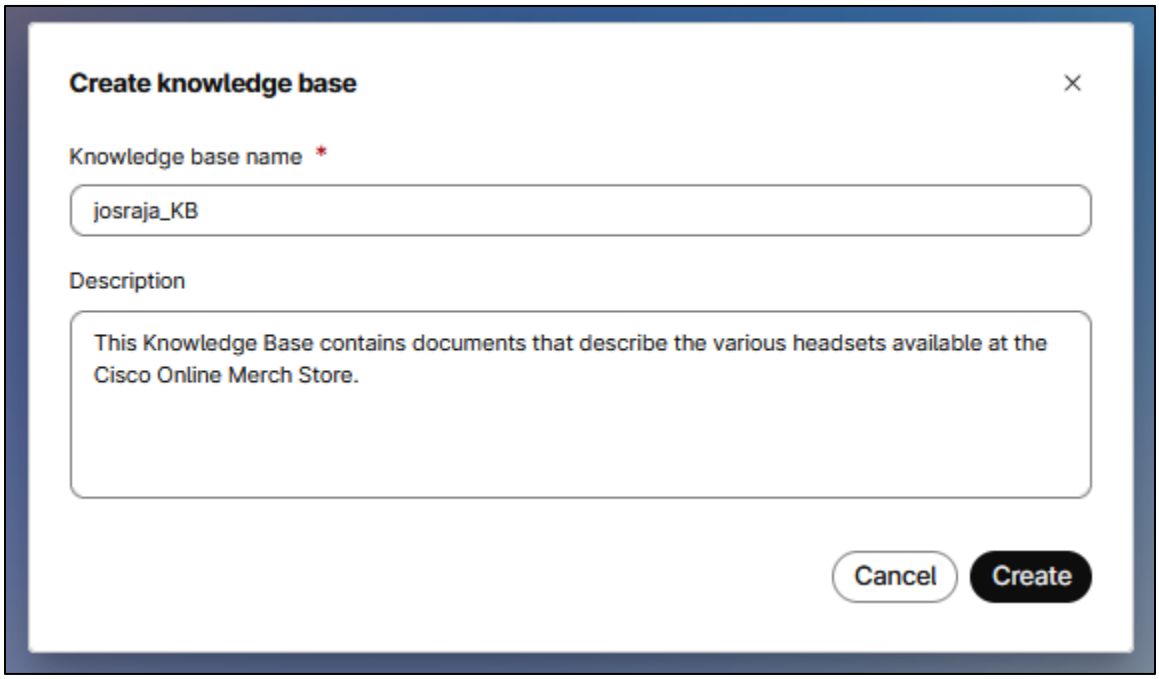
a. Log into Webex Control Hub and navigate to:
Contact Center >> ***AI Agents*** >> ***Build Your AI Agents***





b. Click on the ***Knowledge*** tab followed by the ***Create knowledge base*** button

c. Create a Knowledge Base with your CEC, in order to create a KB unique to you.



d. Download the following 3 headset pdf files and upload them to your KB.

- https://www.cisco.com/c/en/us/products/collateral/collaboration-endpoints/headsets/bang-olufsen-980-ds.pdf
- https://www.cisco.com/c/en/us/products/collateral/collaboration-endpoints/headsets/bang-olufsen-950-ds.pdf
- https://www.cisco.com/c/en/us/products/collateral/collaboration-endpoints/headset-720/headset-720-series-ds.pdf

e. Click ***Save***.

Lab 2: Create an Agent

a. On the AI Studio, click on the AI Agents tab and click the ***Create Agent*** button.

b. Select ***Autonomous*** and enter the following details

- **Agent Name:** *<your-CEC>**-AutonomousAgent***
- **Agent's Goal:** Enter in the Agent's goal. You can use the below example or enter in your own idea for the Agent's goal.
  Example: *You are a polite, professional agent who is an expert in Cisco Headsets. You will help answer customer pre-sales questions on Cisco headsets. You can also help place a new order for a Cisco headset. Additionally, you can also help with tracking order status.*

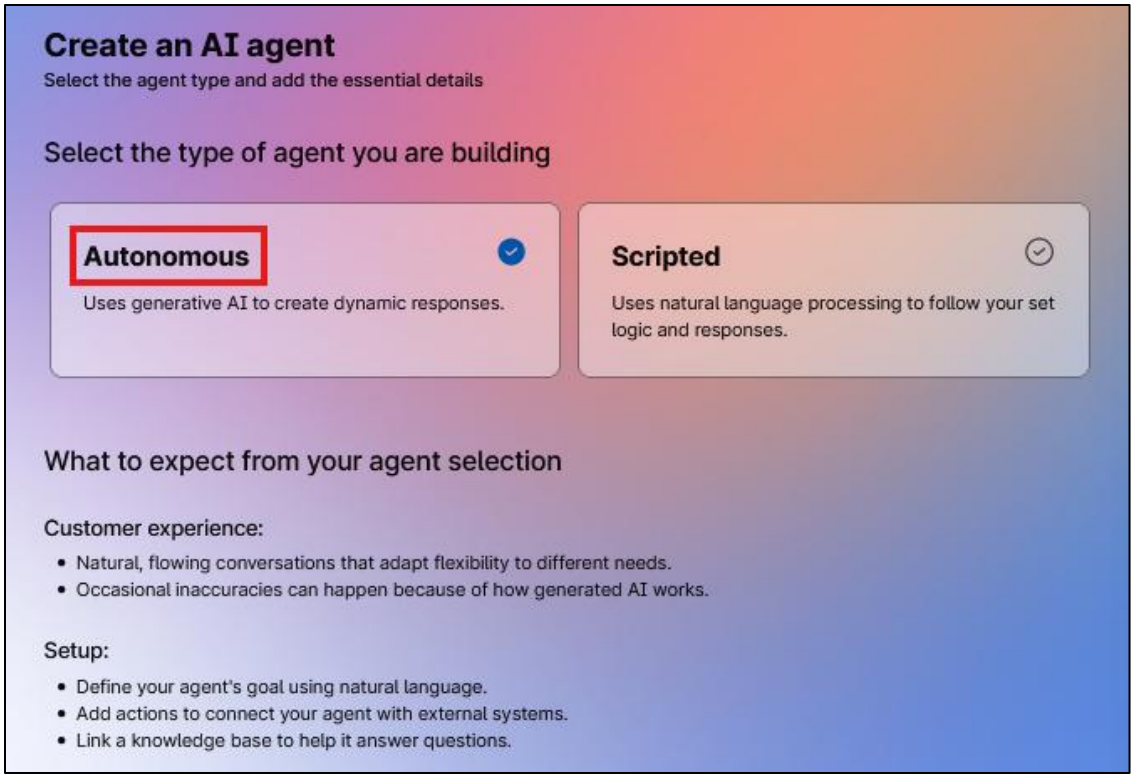
Agent's Goal

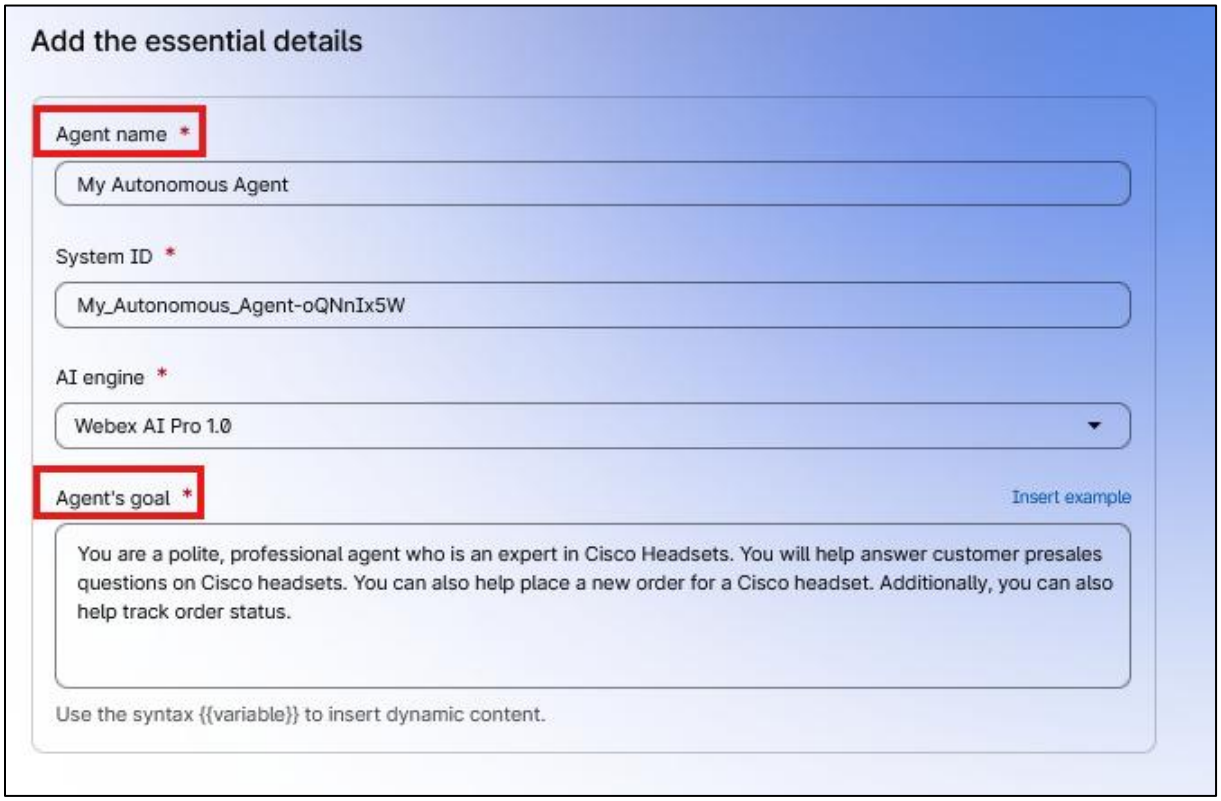
Do's

- Keep the goal short and concise.
- Focus on the overall function or purpose of the AI agent.
- Consider the end-result or benefit for the user.
- Use clear and concise language.
- Ensure the goal aligns with the actions and capabilities of the AI agent.

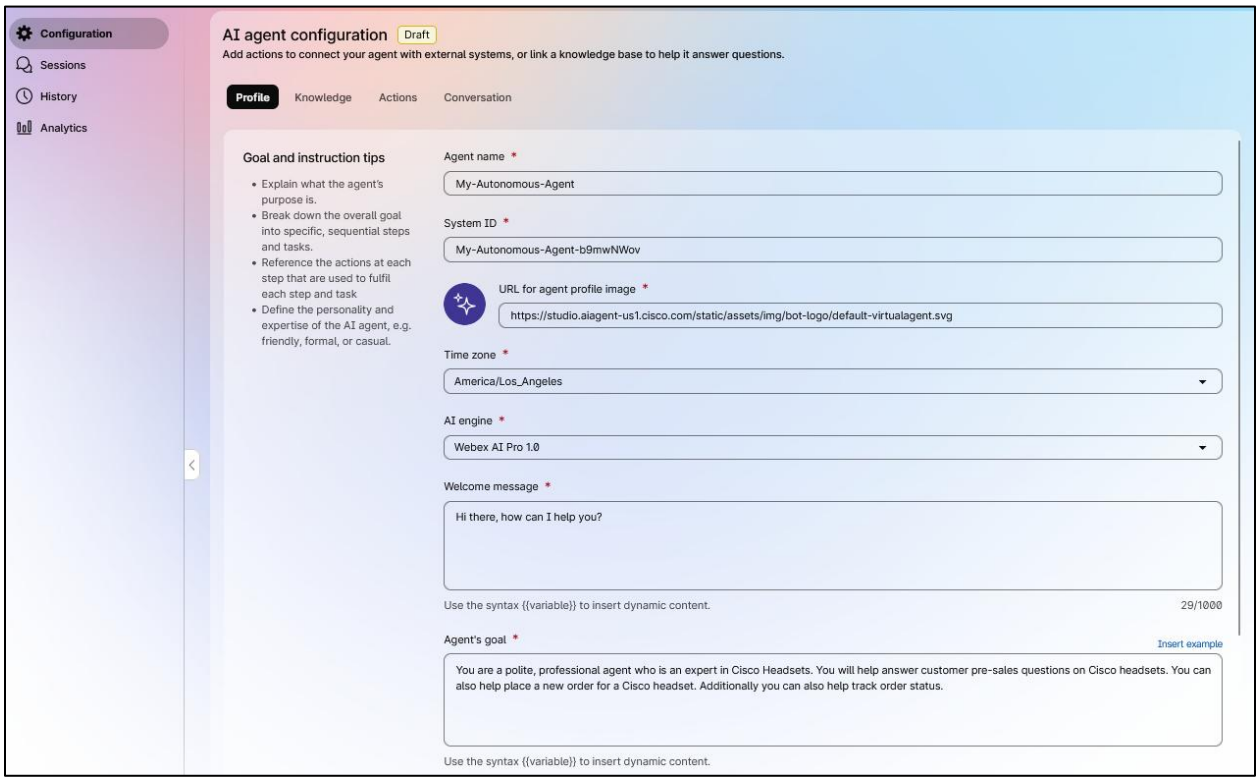
Do Not's

- Do not include specific details like locations, dates, or user information.
- Avoid mentioning particular actions or implementation methods.
- Do not use technical jargon or complex terminology.
- Avoid overly long or complicated goal statements.
- Do not include multiple unrelated goals in a single prompt.
- Avoid using ambiguous or vague language.


Create an AI agent
Select the agent type and add the essential details
Select the type of agent you are building
Autonomous
Uses generative AI to create dynamic responses.
Scripted
Uses natural language processing to follow your set logic and responses.
What to expect from your agent selection
Customer experience:
Natural, flowing conversations that adapt flexibility to different needs.
Occasional inaccuracies can happen because of how generated AI works.
Setup:
Define your agent's goal using natural language.
Add actions to connect your agent with external systems.
Link a knowledge base to help it answer questions.


Add the essential details
Agent name *
My Autonomous Agent
System ID *
My_Autonomous_Agent-oQNnIx5W
AI engine *
Webex AI Pro 1.0
Agent's goal *
Insert example
You are a polite, professional agent who is an expert in Cisco Headsets. You will help answer customer presales questions on Cisco headsets. You can also help place a new order for a Cisco headset. Additionally, you can also help track order status.
Use the syntax {{variable}} to insert dynamic content.

c. Enter the ***Welcome message*** as what you would prefer.



d. Set the instructions for your Agent. You can use the example below as reference.

Example:

"You are an expert customer service representative who can answer any question on the capabilities and features of Cisco headsets. You can also help track the order status of a headset that the caller may have placed before. You should answer all questions in the most professional, helpful way possible.

The customers contacting you are potential customers who will need additional information on the features and capabilities of Cisco headsets in order to make an informed purchasing decision. Or they are customers looking to get an update on the order that was previously placed.

You can also help place a new order for a Cisco headset.

When interacting over text, provide complete answers from the documentation. When interacting over audio channels, focus on providing concise answers while still providing all information needed.

If the customer is not clear about what they ask, clarify their question and ensure that you have all the information you need.

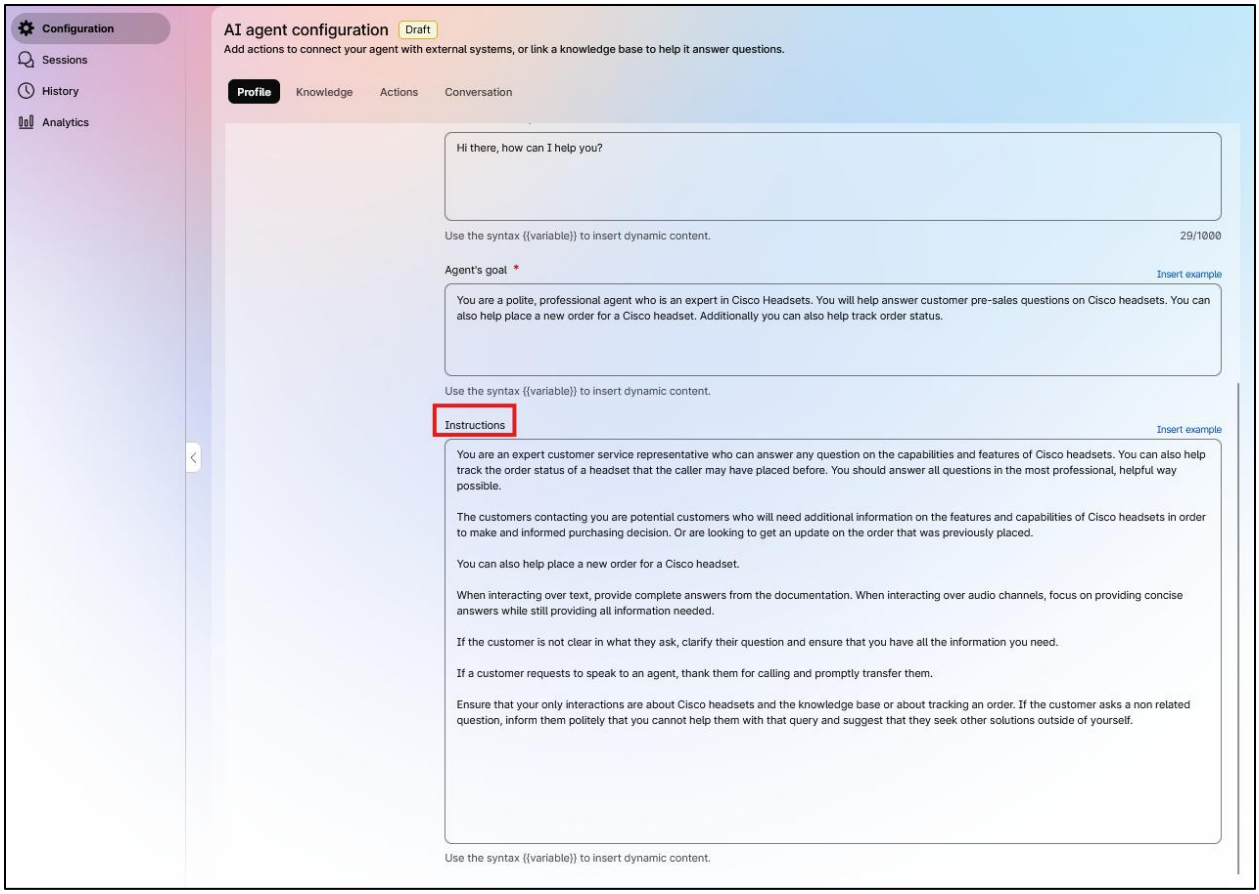
If a customer requests to speak to an agent, thank them for calling and promptly transfer them.

Ensure that your only interactions are about Cisco headsets and the knowledge base or about tracking an order. If the customer asks a non-related question, inform them politely that you cannot help them with that query and suggest that they seek other solutions outside of yourself."

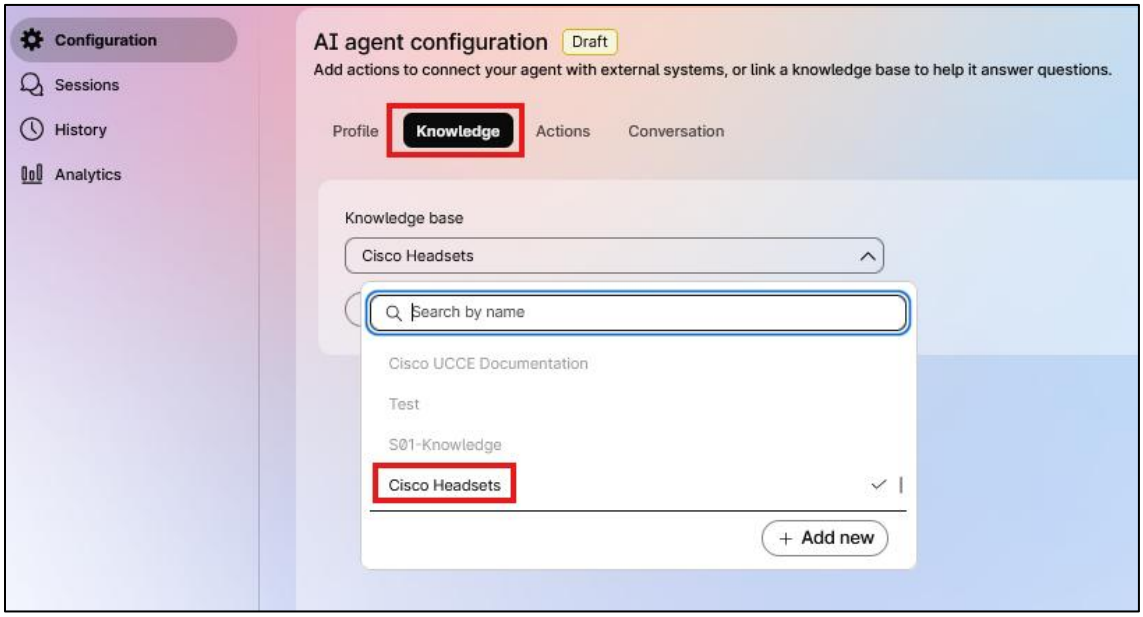
Refer to the following tips when writing instructions for your autonomous AI agents:

- **Keep it simple** — Use clear, concise language. Avoid technical jargon or overly complex sentences.
- **Use markdown** — Use headings and ordered/unordered list markdown for best results.
- **State your AI agent's identity** — Begin by clearly defining the agent's persona (for example "You're a helpful customer support agent…").
- **Break it down** — Outline tasks step by step. For instance, "First, confirm your account number. Then, describe your issue."
- **Plan for errors** — Include fallback phrases such as, "I'm sorry, could you please repeat that?" if the input isn't clear.
- **Preserve context** — Remind the agent to remember previous responses to ensure continuity in long conversations.
- **Reference actions** — Clearly instruct how to use external actions at different steps. Make sure the referenced actions are enabled in the Actions tab to avoid any unexpected behavior.
- **Add guardrails** — Instruct the AI agent to respond only in the context of the goal.
- **Add examples** — To improve accuracy, add examples wherever needed.

For templates on writing instructions refer: Guidelines and best practices for automating with AI agent

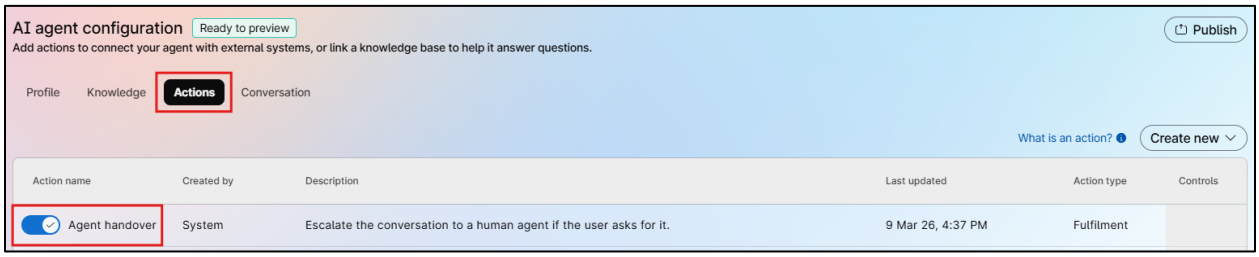


e. Click the ***Knowledge*** tab and from the drop-down menu, select the Knowledge Base you created in Step 1 and then click ***Save***.

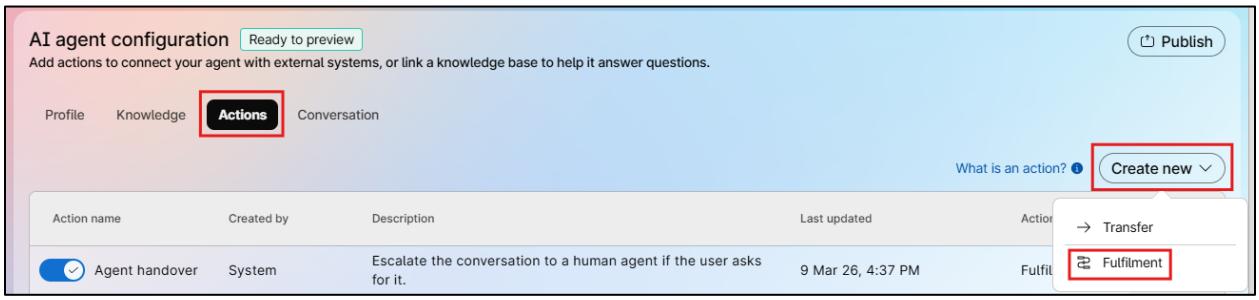


Lab 3: Actions & Fulfillments

a. Select the ***Actions*** tab and ensure the ***Agent handover*** action is enabled. This ensures that if the caller desires to speak to a human agent, the AI agent can transfer the call over to a human agent.



b. Next, click on the ***Create New*** button and select ***Fulfillment***



c. In this Action, we will instruct the AI Agent to collect the 6-digit order number and the 4-digit pin. Once the Action description is provided accordingly, we will now configure the slots (input entities) required to being the task defined in the Action description – which in this case is the 6-digit order number and the 4-digit pin.

- **Action name:** TrackOrder
- **Action description:** *Collect the 6-digit number for the order that is to be tracked and the associated 4-digit pin. And once the fullResp variable is populated, read out the output.*
- **Action scope:** Slot filling and fulfillment

Update action

Enable your AI agent to connect with external systems and perform more complex tasks.

General information

Action name *

TrackOrder

Only alphanumeric, _ and - characters are allowed

Action description *

Collect the 6-digit number for the order that is to be tracked and the associated 4-digit pin. And once the fullResp variable is populated, read out the output.

Use the syntax {{variable}} to insert dynamic content.

Action scope *

Slot filling and fulfillment

Slot filling

Add input entities in a table or define input parameters in JSON for slot filling to ensure the AI agent gathers all necessary information.

Input entities

Use JSON instead

+ New input entity

Entity Name	Type	Value	Description	Example	Required	Controls
packageNum	digits	6	6-digit package number	-	Yes	
pinNum	digits	4	4-digit pin number	-	Yes	

Slot filling - packageNum

Add a new input entity

Specify the details to collect from customers for more accurate AI responses

Entity name *

packageNum

Entity type *

Digit

Entity description *

6-digit package number

Use the syntax {{variable}} to insert dynamic content.

Value *

6

Entity examples

+ Add

Settings

Required

Cancel Add

Slot filling – pinNum

Add a new input entity

Specify the details to collect from customers for more accurate AI responses

Entity name *

pinNum

Entity type *

Digit

Entity description *

4-digit pin number

Use the syntax {{variable}} to insert dynamic content.

Value *

4

Entity examples

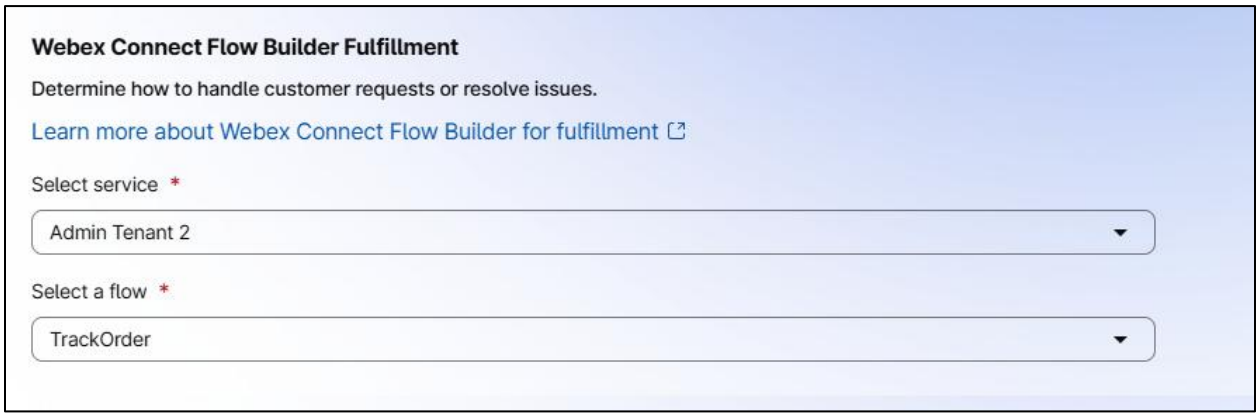
+ Add

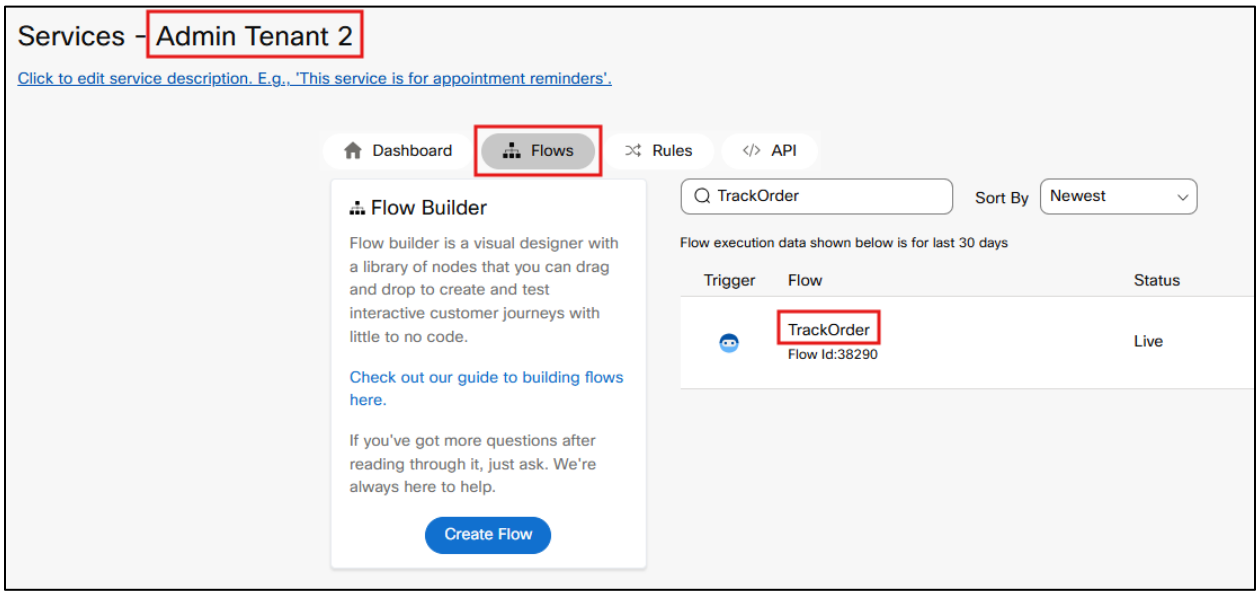
Settings

Required

Cancel Add

d. Now, for the fulfillment of the Action, we need to leverage the capabilities of Webex Connect flows specifically the Task flows. For this lab guide, each tenant will have the Webex Connect flows pre-configured.

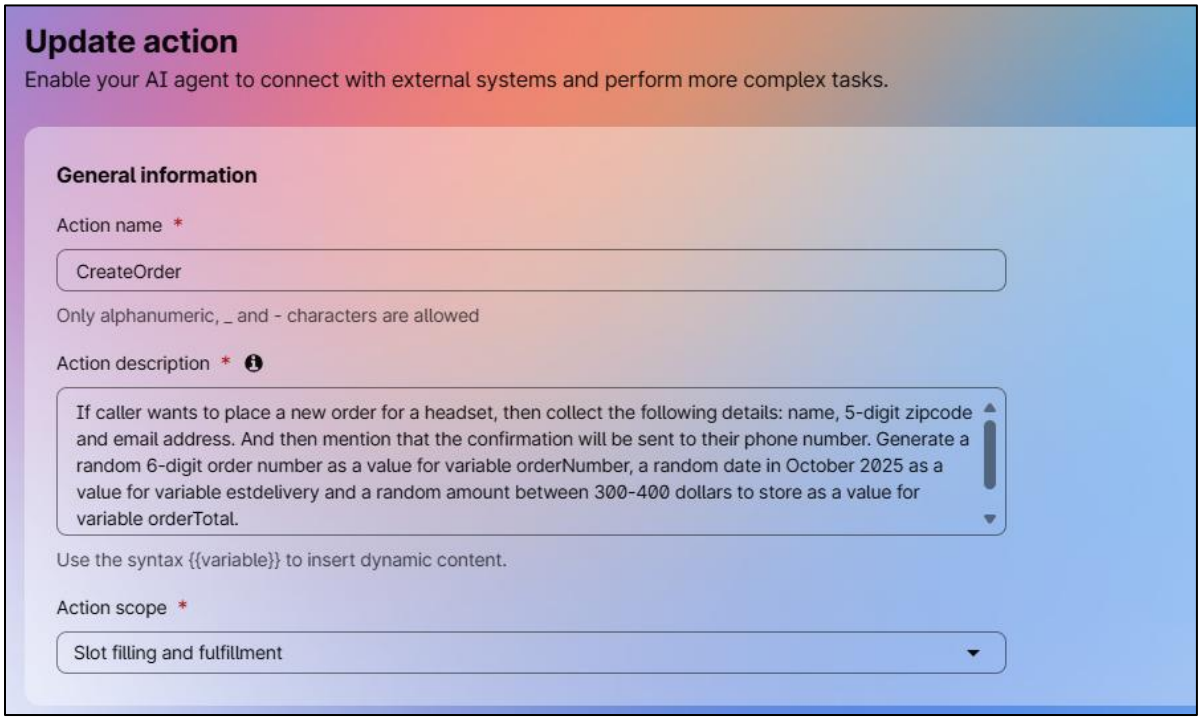


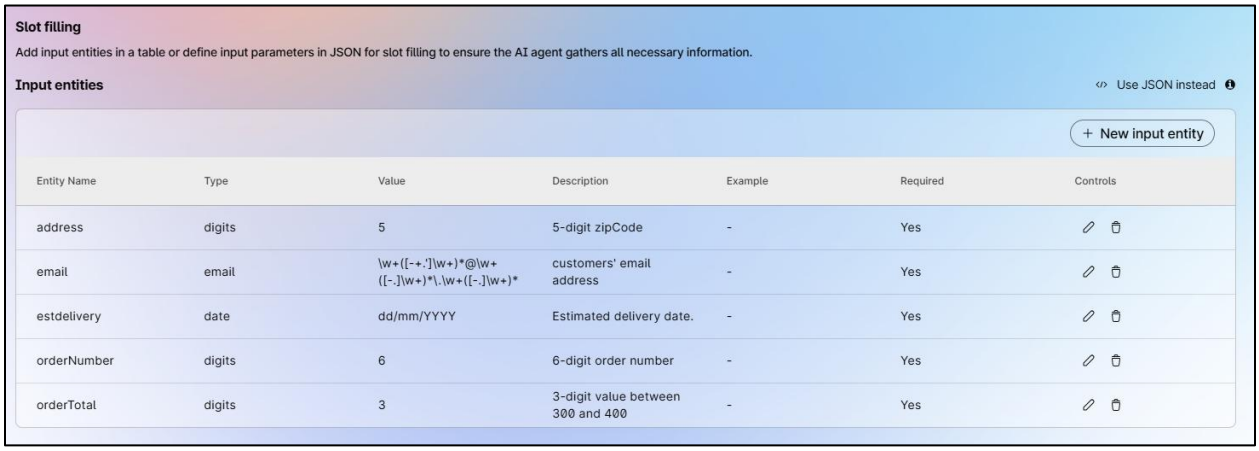


e. Click ***Save***.

f. In this next step, we will now configure an Action called ***CreateOrder*** Action that is executed to place an order for a new headset. The example below is a very trivial example, for the purpose of an easy demonstration. In actuality, the Action to configure a CreateOrder action will be a little more complex. Since the purpose of this lab is to give an overview of Actions & Fulfillments, we are using this trivial example with some pre-set parameters.

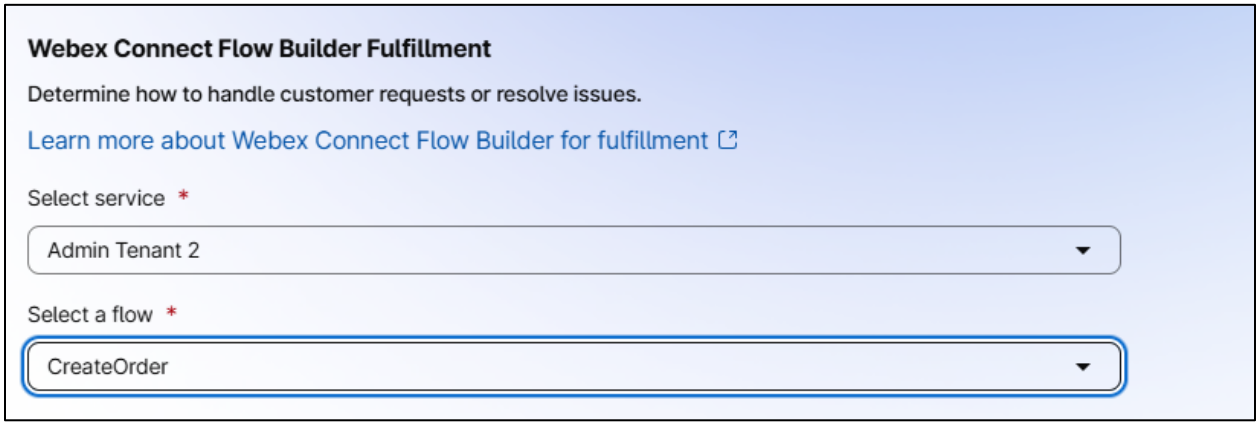
- **Action name:** CreateOrder
- **Action description:** *If caller wants to place a new order for a headset, then collect the following details:* ***name, 5-digit zipcode and email address****. And then mention that the confirmation will be sent to their phone number. Generate a random 6-digit order number as a value for variable orderNumber, a random date in October 2025 as a value for variable estdelivery and a random amount between 300-400 dollars to store as a value for variable orderTotal. Thank the caller for placing the order and ask if they need anything else.*
- **Action scope:** Slot filling and fulfillment

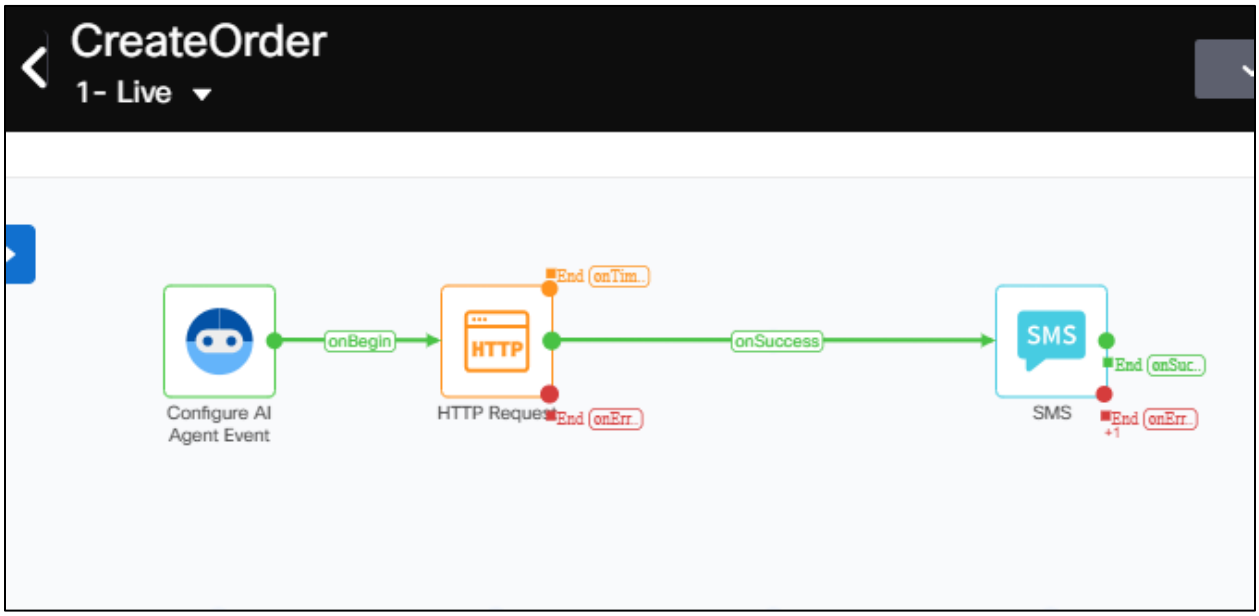




Entity Name	Type	Value	Description	Example	Required	Controls
address	digits	5	5-digit zipCode	-	Yes	
email	email	\w+([-+.']\w+)*@\w+([-.]\w+)*\.\w+([-.]\w+)*	customers' email address	-	Yes	
estdelivery	date	dd/mm/YYYY	Estimated delivery date.	-	Yes	
orderNumber	digits	6	6-digit order number	-	Yes	
orderTotal	digits	3	3-digit value between 300 and 400	-	Yes	

g. Set the following Flow to be used for fulfillment – linked to the CreateOrder Flow on Webex Connect.



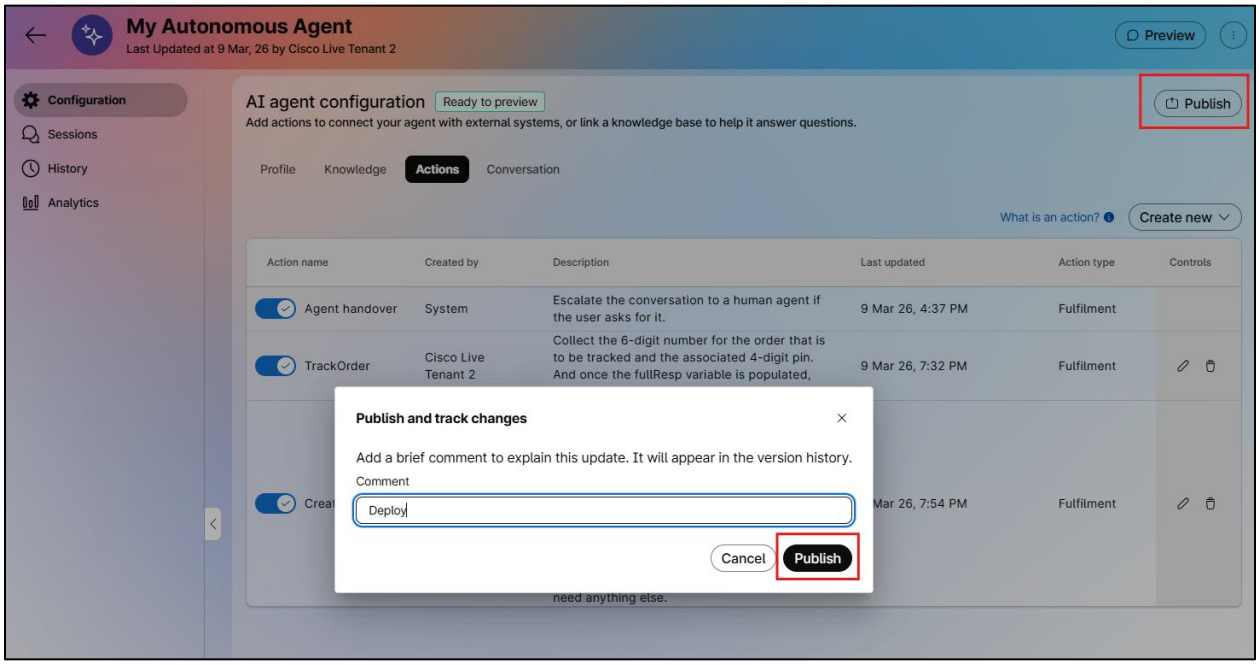


This flow collects the data collected by the AI Agent (name, zipcode and email address) and then makes an API request to an external entity (in this use-case – an order processing system). Once this is executed successfully, the flow then sends a confirmation of the order to the customer via SMS.

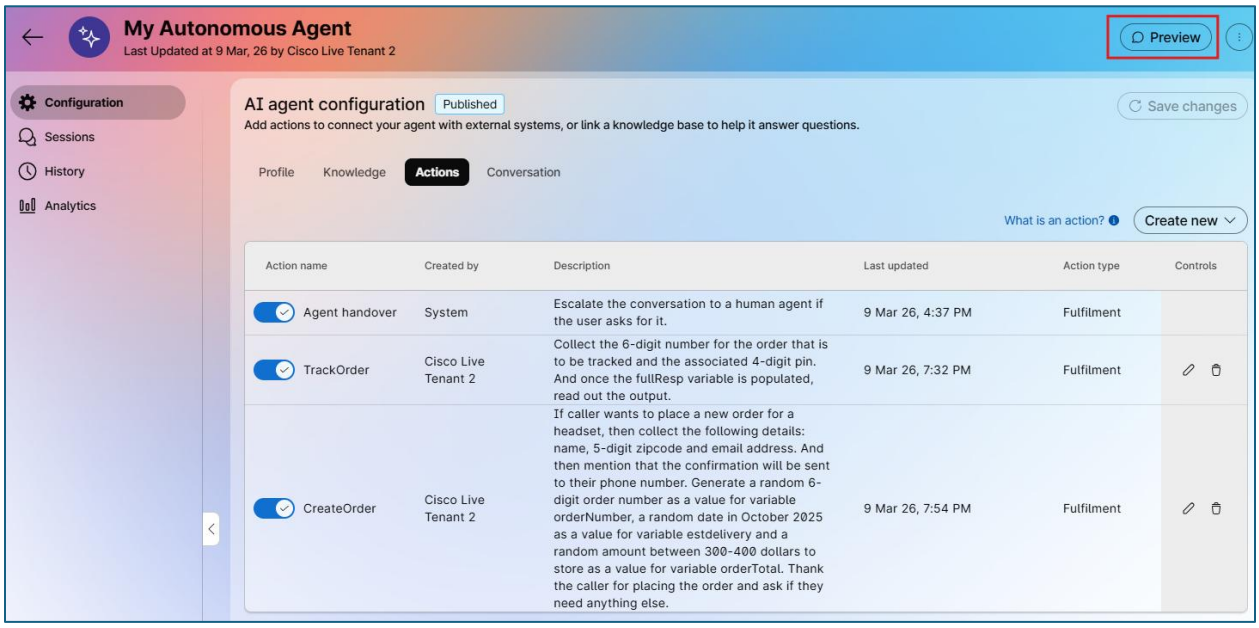
h. Click ***Save***.

Lab 4: Publish your AI Agent

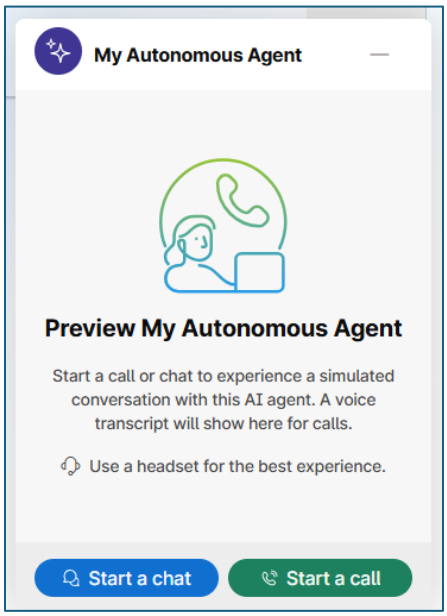
a. Once all the configurations are set and configured, click the ***Publish*** button to have your AI Agent go live!



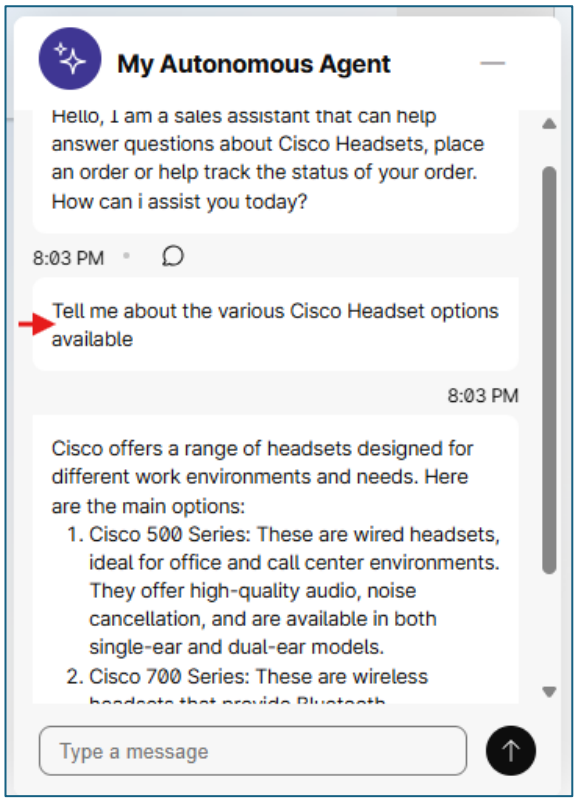
b. Once successfully published, you can test your AI Agent using the ***Preview*** button.



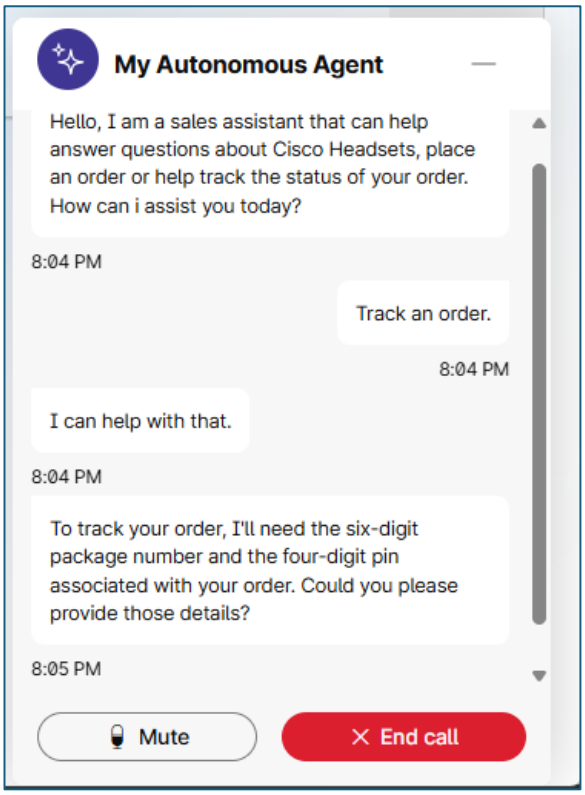
c. You can now test your AI Agent via chat or via call.



Start a chat option

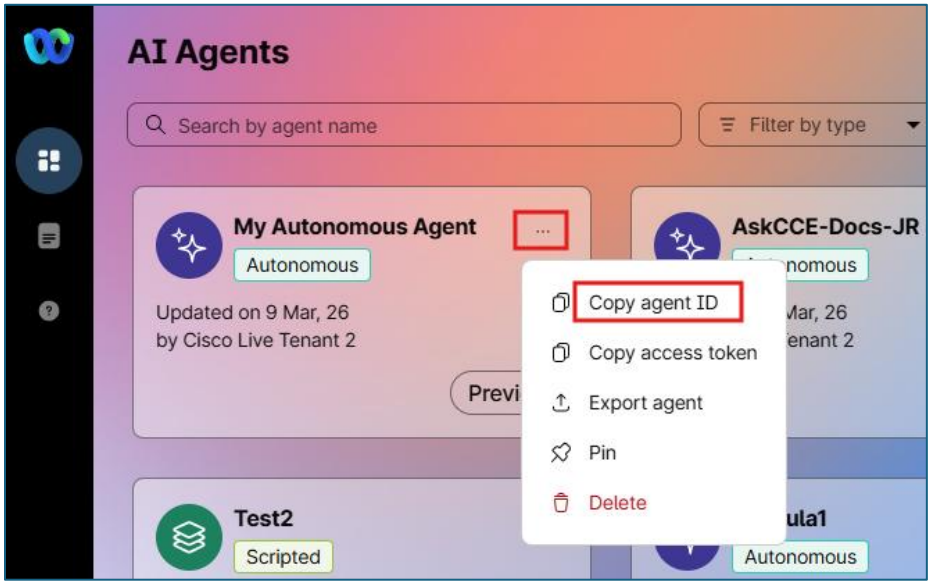


Start a call option

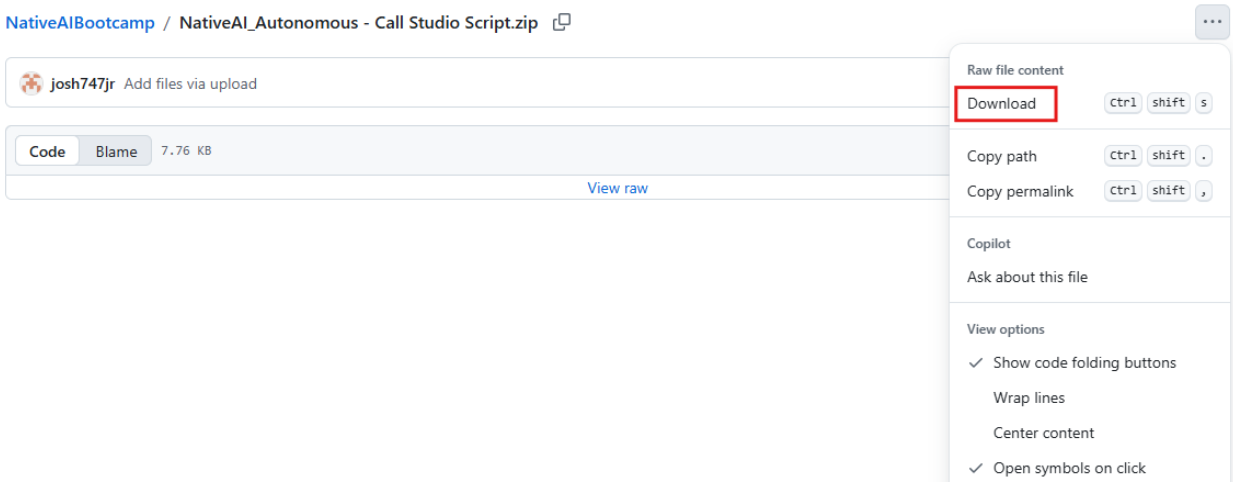


Lab 5: Integrate your AI Agent with your CCE solution

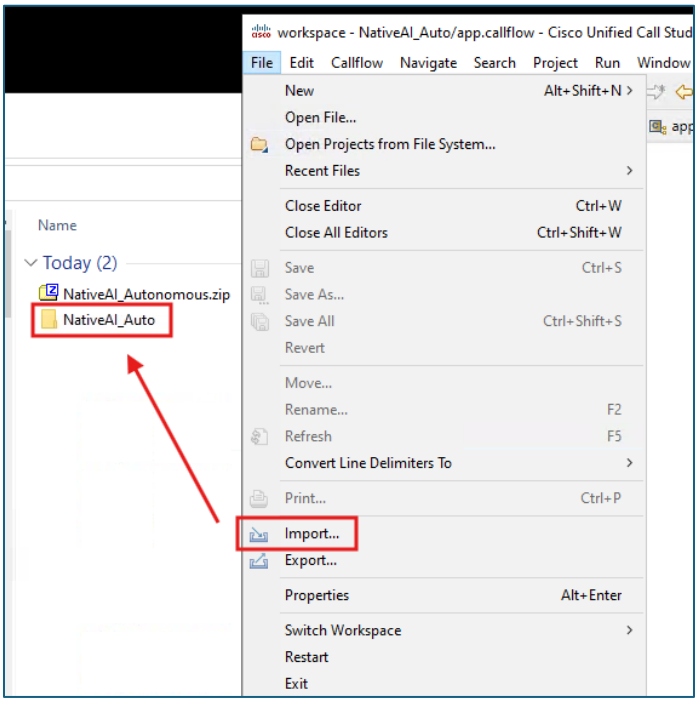
a. From your AI Agent, copy the agent ID and save it to a text file.



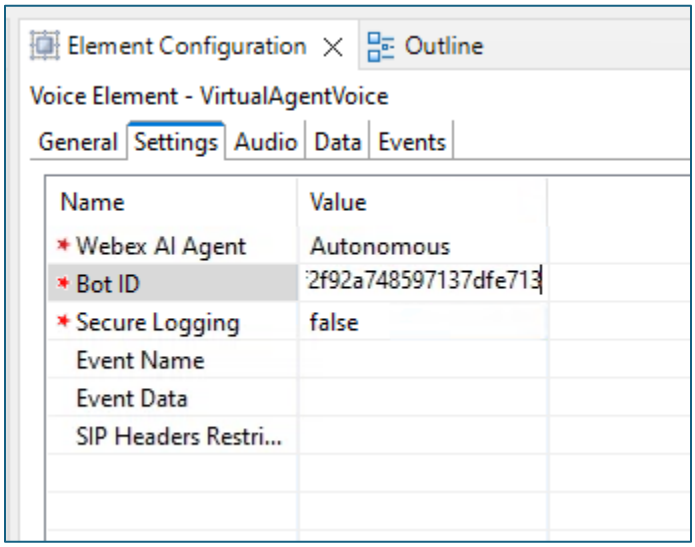
b. On WKSTN1 (in dCloud), open ***mRemoteNG*** from the desktop and RDP into the CVP server. Download the Autonomous AI Call Studio script from the following link: [NativeAI_Auto]

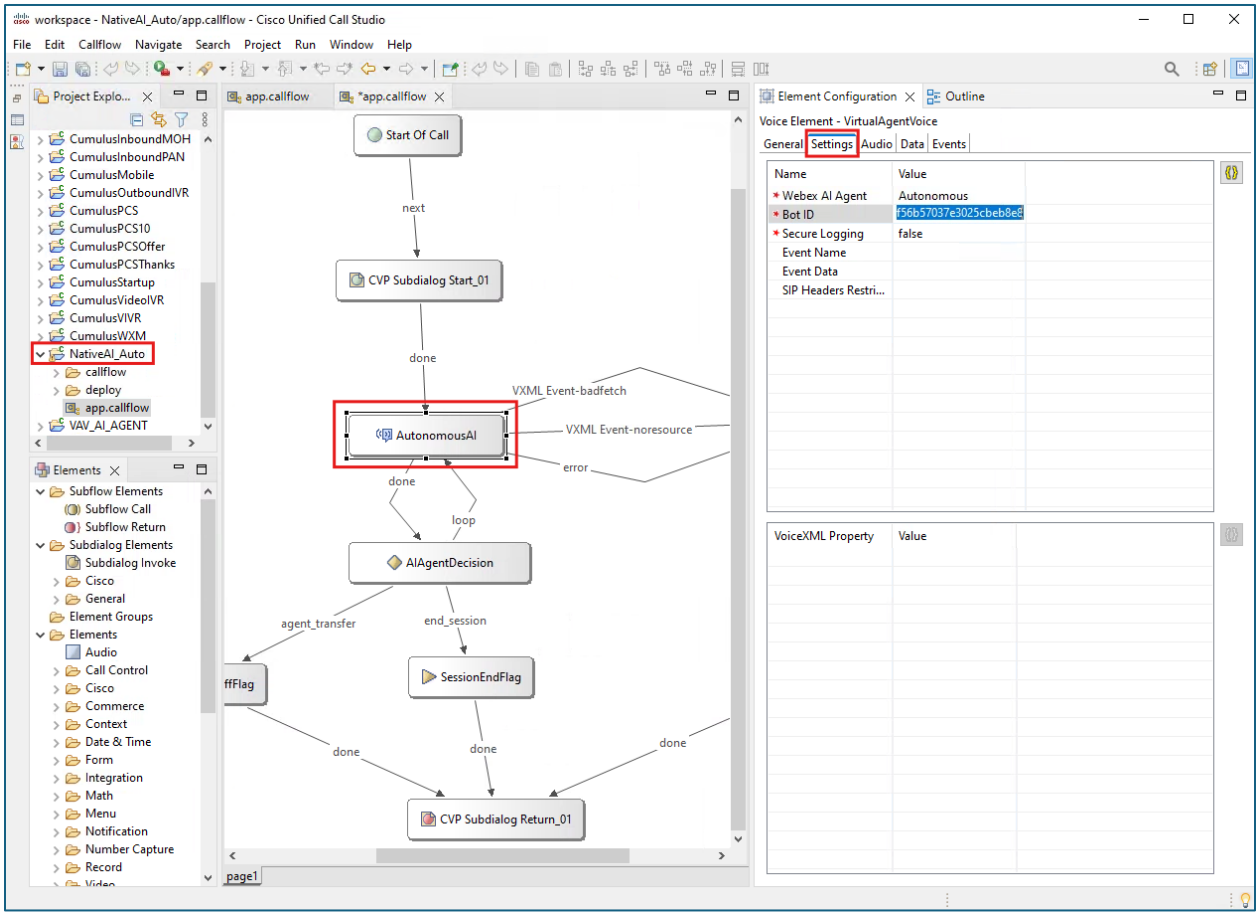


c. Open the Call Studio app and import the ***NativeAI_Auto*** script (unzip the downloaded folder first)

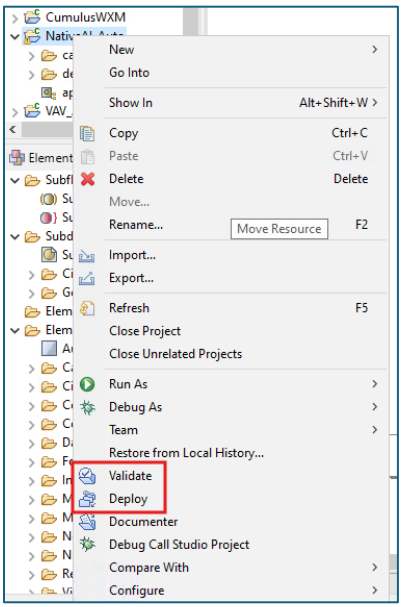


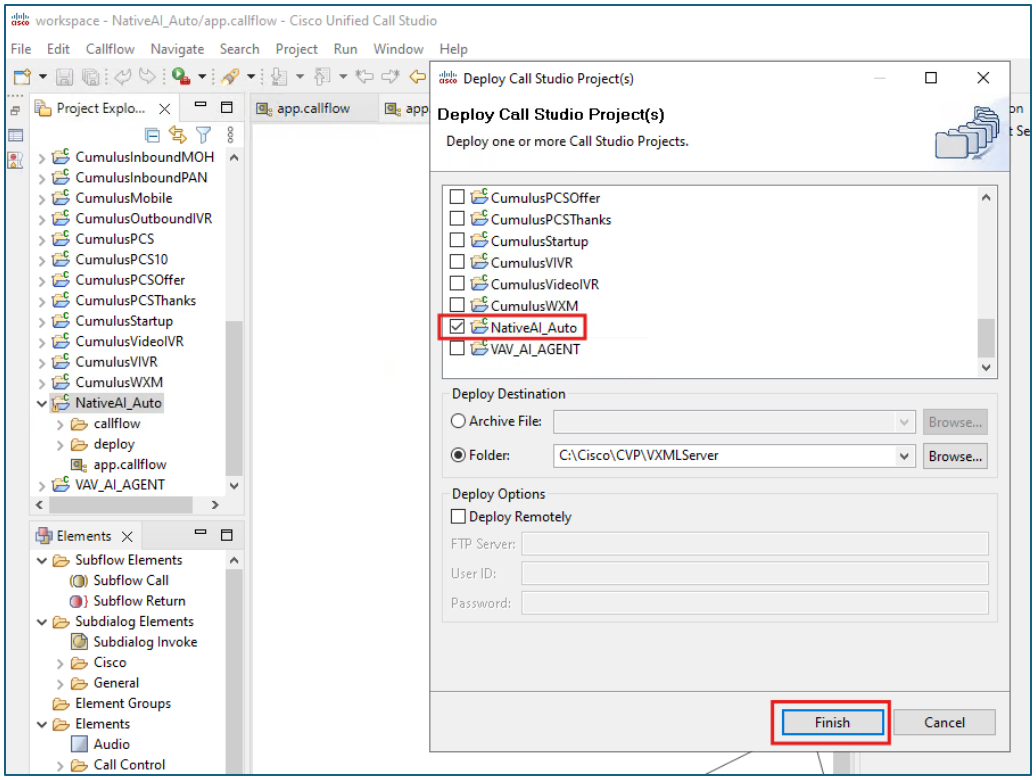
d. Once imported, select the VirtualAgentVoice (VAV) element named as AutonomousAI >> select the Settings tab and set the below config:
Webex AI Agent: Autonomous
Bot ID: *<your-agent-ID>*



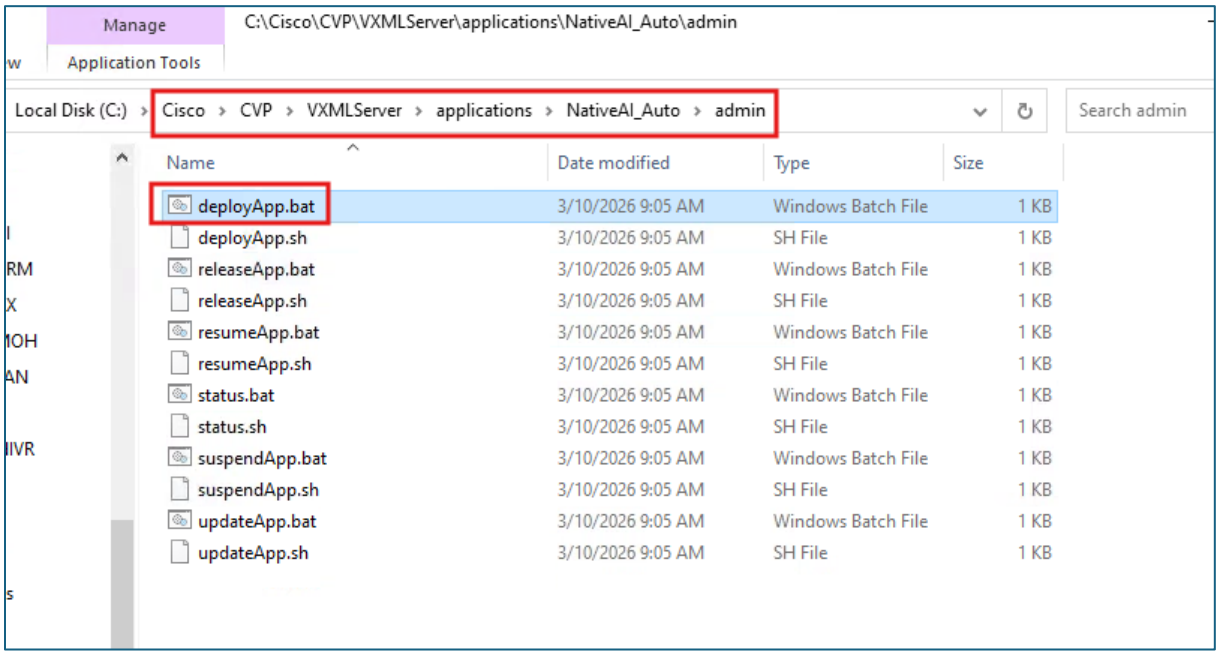


e. Enter crtl+S to save the changes made. Right click on the NativeAI_Auto app and click ***Validate*** >> Next, right-click and click ***Deploy***.

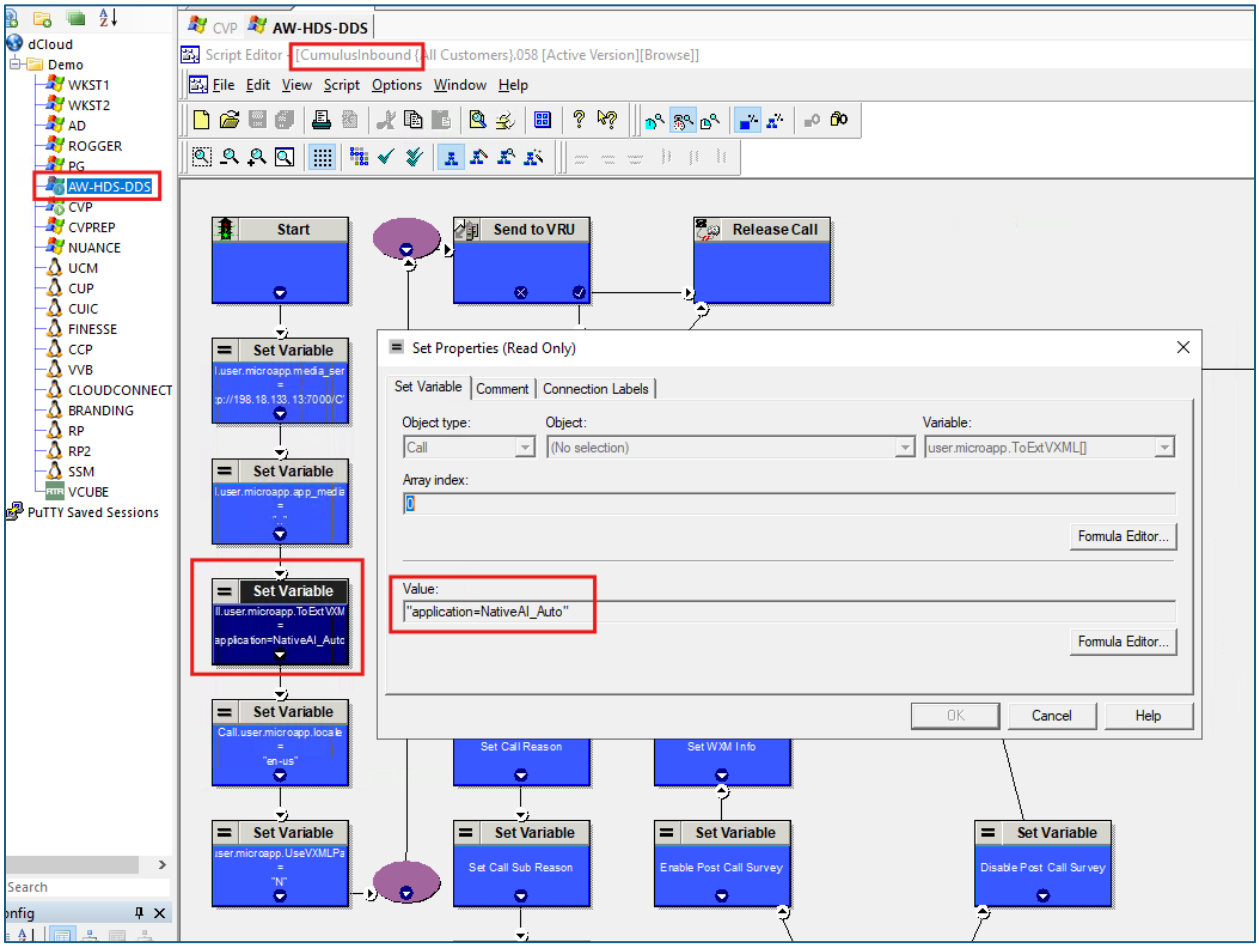




f. To complete deploying the app, navigate to ***Cisco\CVP\VXMLServer\applications\NativeAI_Auto\admin*** and double-click the ***deployApp.bat*** file. Enter y or yes, when prompter



g. Lastly, using ***mRemoteNG***, RDP into the ***AW-HDS-DDS*** server >> open Script Editor >> open the script named ***CumulusInbound*** >> enter Edit mode >> modify the Set properties node for ToExtVXML[] and set it to ***application=NativeAI_Auto*** >> Save.



Lab 6: Set traces and place a test call

Task 1. Setting traces on the SpeechServer

a. On Chrome browser (in dCloud), enter the URL: ***https://vvb1.dcloud.cisco.com***

Select ***Cisco VVB Serviceability*** from the drop down to the right and click ***Go***. Login with the following credentials:
Username: Administrator
Password: dCloud!23

b. Next, navigate to ***Trace*** >> ***Configuration*** >> select ***Speech Server*** from the drop down and click ***Go***.
and set trace level to "***XDebugging3***" by checking all 4 boxes >> click ***Save***.

Task 2. Place a test call and note down the time

Note: You will need to use your Webex App (with the Cisco Extension) to call into the number.

c. On WKST1 (in dCloud), open a browser and open a new tab, then in the default page that appears, select ***Demo Links*** >> ***Demo Website***.

d. In the Cumulus Finance website that is shown, select the blue box on the right-hand side that reads "***Talk to an Expert***".
In the box that pops out, select the ***Call Us*** link. In the box that pops up, note the ***Main*** number, this is what you will use to test your lab.

IMPORTANT: THE NUMBER SHOWN IN THE IMAGE IS NOT THE NUMBER YOU WILL USE FOR YOUR LAB. ENSURE THAT YOU FIND THE NUMBER FOR YOUR SESSION!

e. Use your Webex App to call the number. You should hear the bot greet you by name and then ask how it can help.

f. Test your bot for the different tasks and actions configured.

Sample utterances for the test call:

Test 1. Tell me about the different Cisco headset options available.

Test 2. Track an order (when prompted, respond with a random 6-digit order number and random 4-digit pin)

Test 3. I want to place an order for headset, (when prompted provide a random name, a random 5-digit zip code and a random email address)

Step 7: Troubleshooting Basics

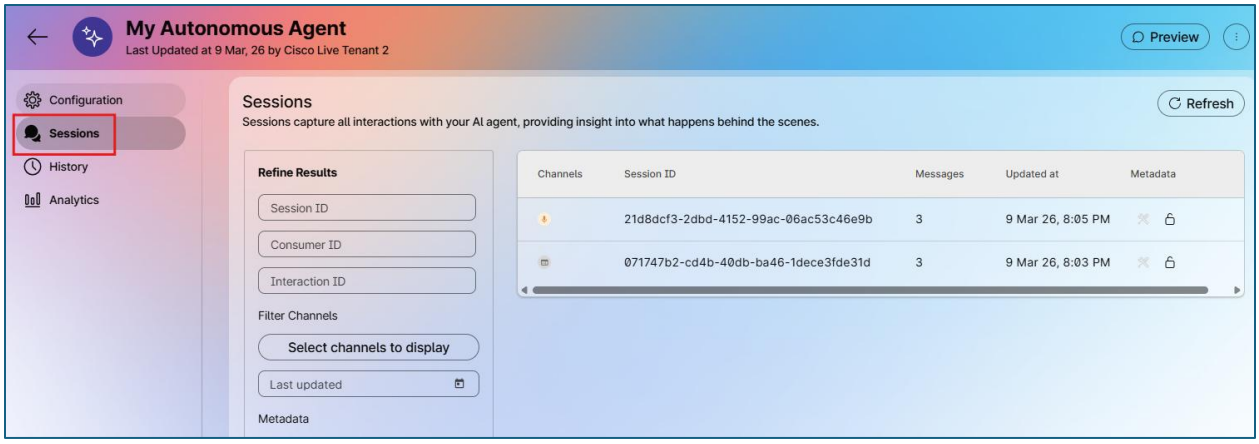
In order review the logs or to troubleshoot any issues, the following logs are available:

- AI Agent logs
- CCE logs (ICM and CVP)

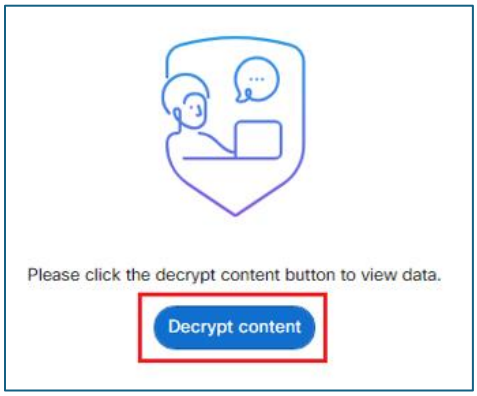
Step 1. AI Agent logs:

a. Navigate back to AI Agent dashboard and select your AI Agent.

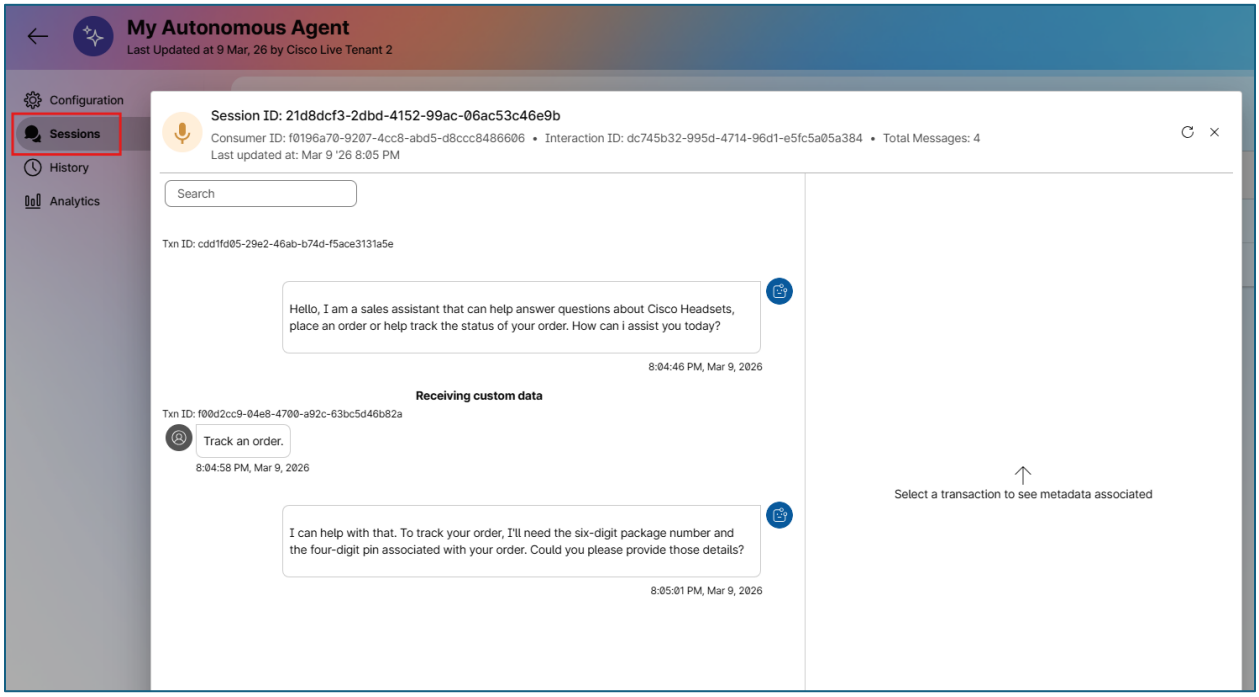
b. From the left side menu, click on ***Sessions*** >> find the session that matches the timestamp that you placed the call and click on it.

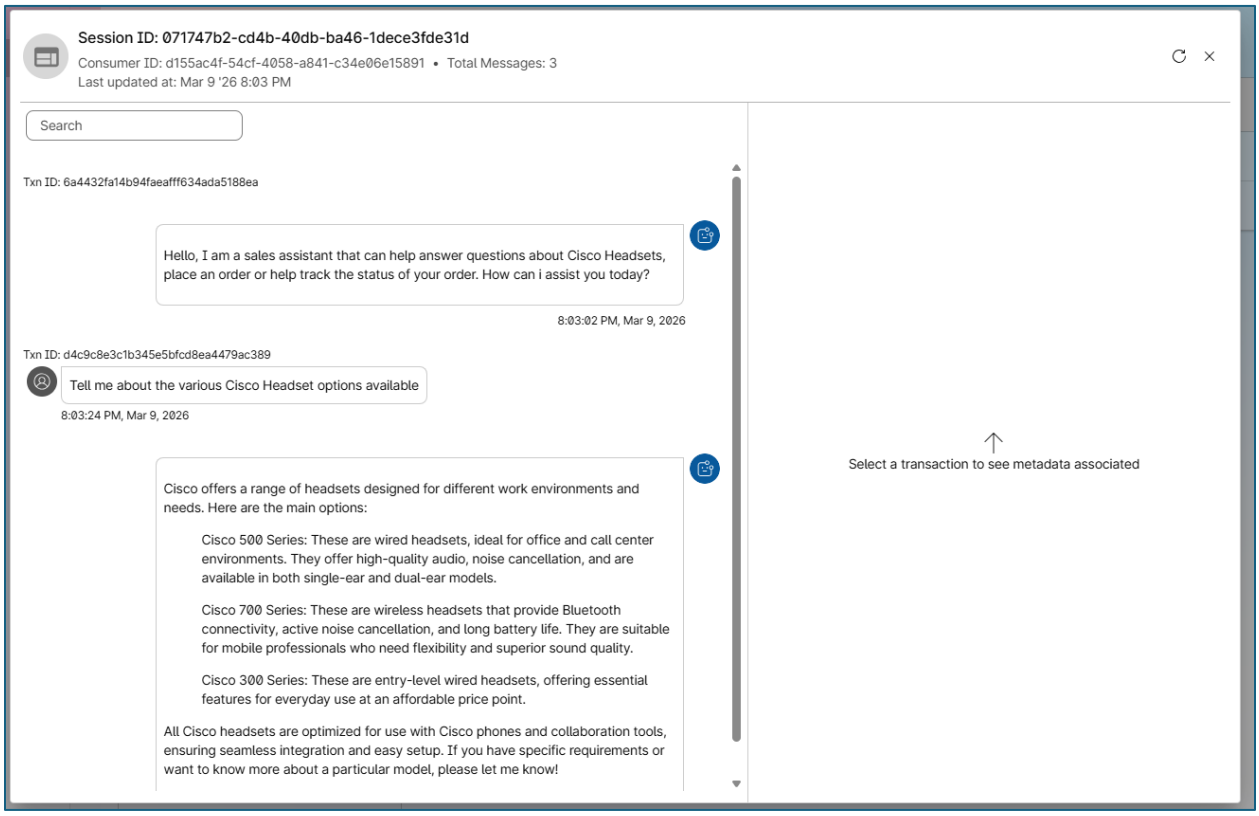


c. Next, click on ***Decrypt content***.



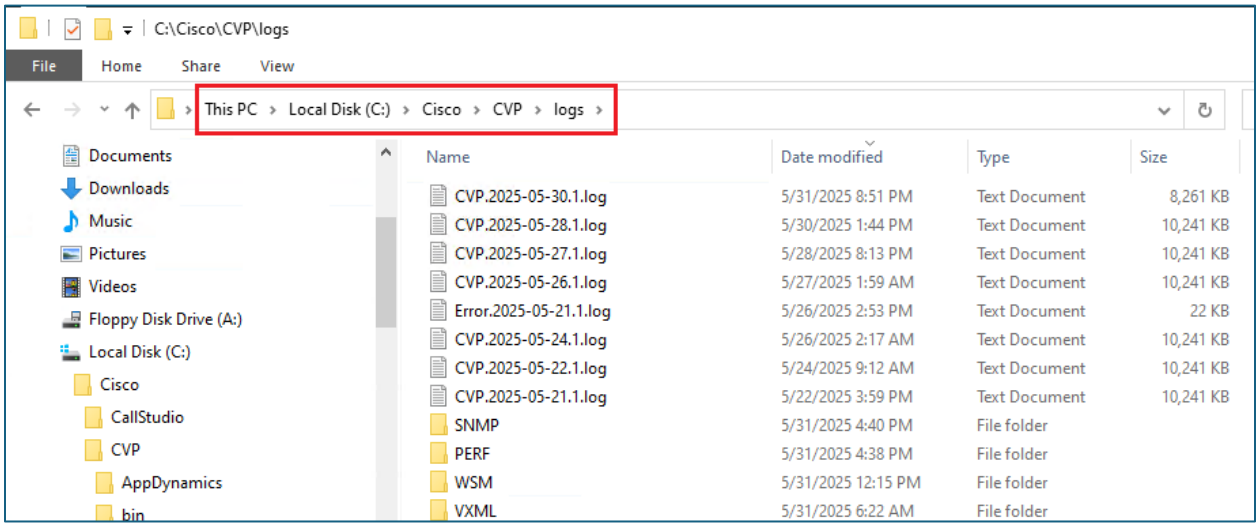
d. This will now open up the entire session for the call and the associated metadata for each of the transactions.

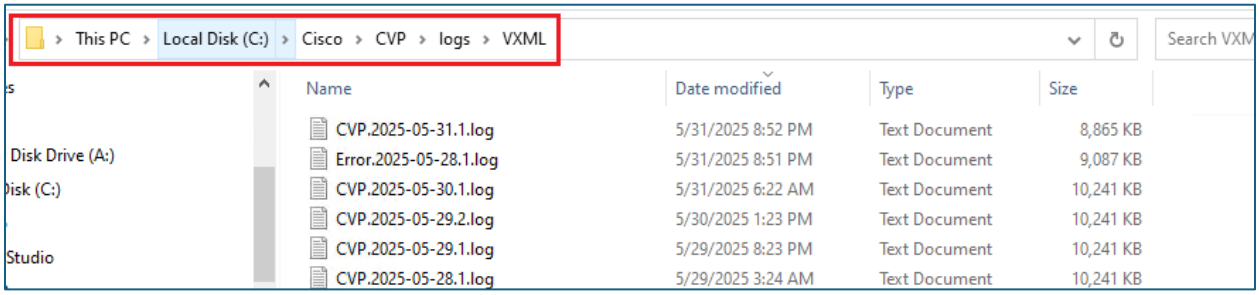


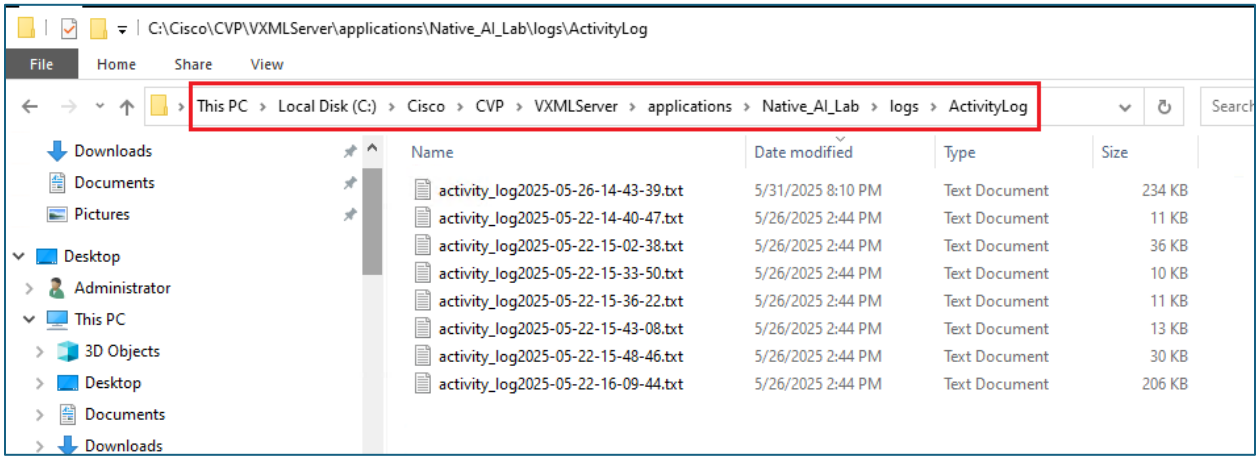


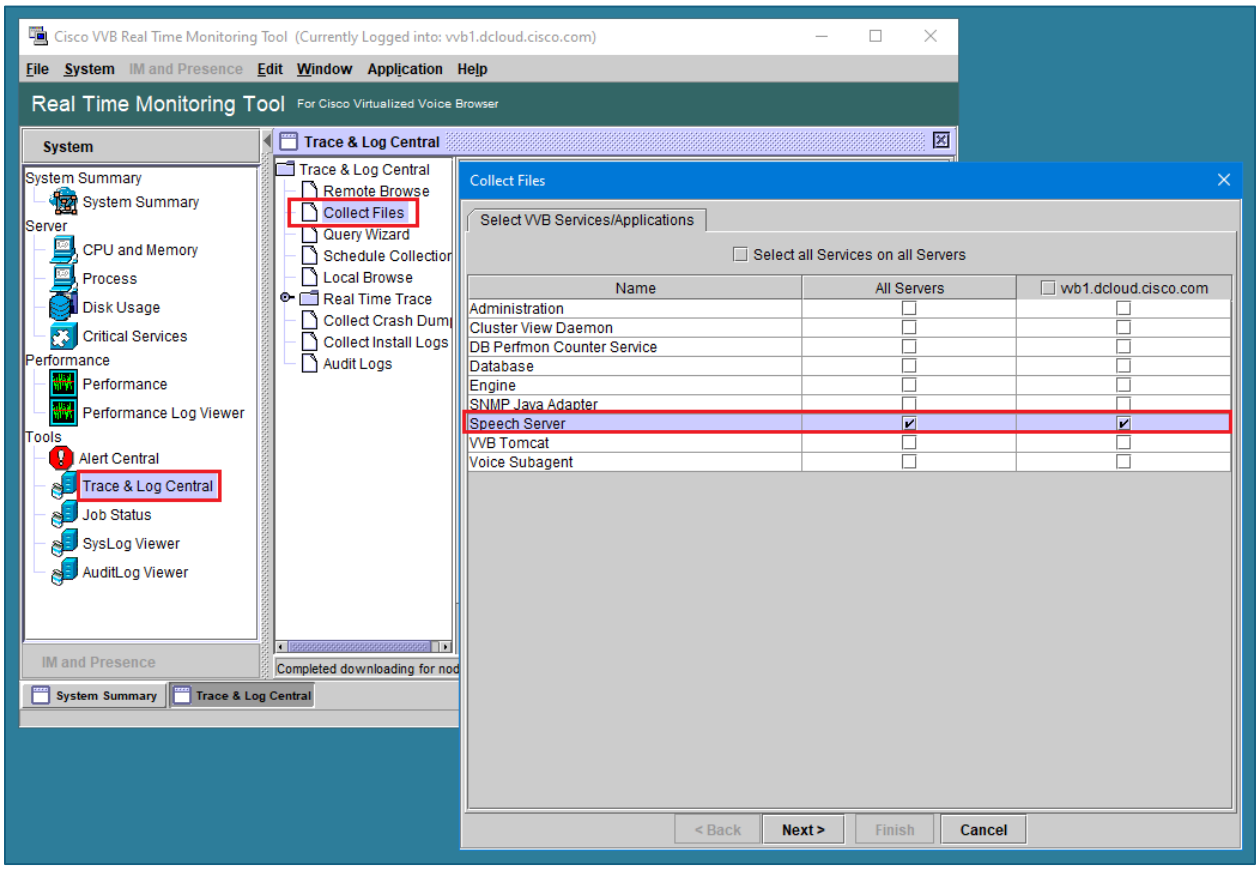
Step 2. CCE side:

- CVP Call Server logs
- CVP VXML Server
- Activity Logs
- SpeechServer








Cisco VVB Real Time Monitoring Tool (Currently Logged into: vvb1.dcloud.cisco.com)
File System IM and Presence Edit Window Application Help
Real Time Monitoring Tool For Cisco Virtualized Voice Browser
System
System Summary
System Summary
Server
CPU and Memory
Process
Disk Usage
Critical Services
Performance
Performance
Performance Log Viewer
Tools
Alert Central
Trace & Log Central
Job Status
SysLog Viewer
AuditLog Viewer
IM and Presence
Trace & Log Central
Trace & Log Central
Remote Browse
Collect Files
Query Wizard
Schedule Collectior
Local Browse
Real Time Trace
Collect Crash Dum
Collect Install Logs
Audit Logs
Completed downloading for nod
System Summary
Trace & Log Central
Collect Files
Select VVB Services/Applications
Select all Services on all Servers
Name
All Servers
vvb1.dcloud.cisco.com
Administration
Cluster View Daemon
DB Perfmon Counter Service
Database
Engine
SNMP Java Adapter
Speech Server
VVB Tomcat
Voice Subagent
< Back
Next >
Finish
Cancel